

The Depth of a Bent Partition Is a Power of the Characteristic

Abstract

Let p be a prime and let $n > 0$ be even. We prove, without any regularity or structural hypothesis, that the depth of every bent partition of $\mathbb{F}_p^n$ is a power of p . The proof averages, over all balanced labelings of the blocks, the number of zero values of one fixed nonzero derivative.

1 Definitions and exact statement

Put $V = \mathbb{F}_p^n$, equip V with the standard $\mathbb{F}_p$ -valued inner product, and set

$$\zeta_p = \exp(2\pi i/p).$$

For a function $f : V \rightarrow \mathbb{F}_p$, its Walsh transform is

$$W_f(b) = \sum_{x \in V} \zeta_p^{f(x) - \langle b, x \rangle} \quad (b \in V).$$

The function f is *bent* if

$$|W_f(b)| = p^{n/2} \quad \text{for every } b \in V.$$

Let $\Omega = \{A_1, \dots, A_K\}$ be a partition of V into nonempty sets, and suppose that $p \mid K$. A labeling $\ell : \{1, \dots, K\} \rightarrow \mathbb{F}_p$ is called *balanced* if

$$|\ell^{-1}(c)| = K/p \quad \text{for every } c \in \mathbb{F}_p.$$

It induces the function $f_\ell : V \rightarrow \mathbb{F}_p$ defined by

$$f_\ell(x) = \ell(i) \quad \text{when } x \in A_i.$$

The partition Ω is a *bent partition of depth K* if f_ℓ is bent for every balanced labeling ℓ .

Theorem 1 (Exact depth theorem). *Let p be a prime, let $n > 0$ be even, and let $\Omega = \{A_1, \dots, A_K\}$ be a bent partition of $\mathbb{F}_p^n$ of depth K in the sense defined above. Then there is an integer k with*

$$1 \leq k \leq n$$

such that

$$K = p^k.$$

No hypothesis is imposed on regularity, weak regularity, the sizes or types of the blocks, normality, spread structure, vectorial-dual-bent structure, or the manner in which the partition was constructed.

2 A derivative lemma

For $a \in V$, define

$$D_a f(x) = f(x + a) - f(x)$$

and the autocorrelation

$$C_f(a) = \sum_{x \in V} \zeta_p^{D_a f(x)}.$$

Lemma 2. *If $f : V \rightarrow \mathbb{F}_p$ is bent and $a \neq 0$, then $D_a f$ is balanced. More precisely, for every $c \in \mathbb{F}_p$,*

$$\#\{x \in V : D_a f(x) = c\} = p^{n-1}.$$

Proof. Writing $a = x - y$ in the product $W_f(b) \overline{W_f(b)}$ gives the finite Fourier identity

$$|W_f(b)|^2 = \sum_{a \in V} C_f(a) \zeta_p^{-\langle b, a \rangle}. \quad (1)$$

Since f is bent, the left-hand side of (1) is the constant p^n . Fourier inversion on the additive group of V therefore gives

$$C_f(a) = 0 \quad (a \neq 0).$$

Fix $a \neq 0$, and for $c \in \mathbb{F}_p$ put

$$m_c = \#\{x \in V : D_a f(x) = c\}.$$

After identifying $\mathbb{F}_p$ with $\{0, 1, \dots, p-1\}$ in exponents, the preceding equality says

$$\sum_{c=0}^{p-1} m_c \zeta_p^c = 0.$$

Thus the polynomial $P(X) = \sum_{c=0}^{p-1} m_c X^c \in \mathbb{Z}[X]$ vanishes at ζ_p . Because p is prime, the minimal polynomial of ζ_p over $\mathbb{Q}$ is

$$\Phi_p(X) = 1 + X + \cdots + X^{p-1}.$$

As $\deg P \leq p-1 = \deg \Phi_p$, it follows that $P(X) = t\Phi_p(X)$ for some rational number t . Hence all the m_c are equal. Their sum is $|V| = p^n$, so $m_c = p^n/p = p^{n-1}$ for every $c \in \mathbb{F}_p$. $\square$

3 A balanced-labeling lemma

Lemma 3. *Let $K = pr$, and choose uniformly at random a labeling $\ell : \{1, \dots, K\} \rightarrow \mathbb{F}_p$ satisfying*

$$|\ell^{-1}(c)| = r \quad \text{for every } c \in \mathbb{F}_p.$$

For any two distinct block indices $i \neq j$,

$$\Pr(\ell(i) = \ell(j)) = \frac{r-1}{K-1}.$$

This probability is independent of the sizes of the blocks and of the choice of the distinct indices i, j ; it depends only on K, p, r .

Proof. For every $c \in \mathbb{F}_p$, uniformity of the balanced labelings gives

$$\Pr(\ell(i) = c) = \frac{r}{K} = \frac{1}{p}.$$

Conditional on $\ell(i) = c$, exactly $r-1$ of the remaining $K-1$ block indices have label c . Hence

$$\begin{aligned} \Pr(\ell(i) = \ell(j)) &= \sum_{c \in \mathbb{F}_p} \Pr(\ell(i) = c) \Pr(\ell(j) = c \mid \ell(i) = c) \\ &= p \cdot \frac{1}{p} \cdot \frac{r-1}{K-1} = \frac{r-1}{K-1}. \end{aligned}$$

The calculation uses only the indexed set of K blocks and their label multiplicities, not their cardinalities, and it is identical for every ordered pair of distinct indices i, j . $\square$

4 Proof of the depth theorem

Proof of Theorem 1. Write

$$Q = p^n, \quad r = K/p.$$

Here r is a positive integer. Since $n > 0$, choose and fix $a \in V \setminus \{0\}$. Let

$$T_a = \#\{x \in V : x \text{ and } x + a \text{ belong to the same block of } \Omega\}. \quad (2)$$

In particular, T_a is a nonnegative integer.

Choose a balanced labeling ℓ uniformly at random from the finite set of all balanced labelings. For every such ℓ , Lemma 2, applied to the bent function f_ℓ , shows that

$$\#\{x \in V : D_a f_\ell(x) = 0\} = Q/p. \quad (3)$$

We now compute the expectation of the left-hand side of (3) by considering each $x \in V$. If x and $x + a$ belong to the same block, then $D_a f_\ell(x) = 0$ for every labeling ℓ . If they belong to two distinct blocks, then $D_a f_\ell(x) = 0$ exactly when those two blocks receive the same label. By Lemma 3,

$$\Pr(D_a f_\ell(x) = 0) = \frac{r-1}{K-1}$$

in the distinct-block case. If I_x denotes the indicator of this zero derivative event, then $\mathbb{E}[\sum_x I_x] = \sum_x \mathbb{E}[I_x]$; thus linearity of expectation uses only the marginal probabilities and requires no independence among the events indexed by x . It now yields

$$\frac{Q}{p} = T_a + (Q - T_a) \frac{r-1}{K-1}. \quad (4)$$

Substituting $r = K/p$ into (4) and multiplying by $p(K-1)$ gives

$$Q(K-1) = pT_a(K-1) + (Q - T_a)(K-p).$$

The right-hand side is

$$(p-1)KT_a + QK - pQ.$$

After cancellation, we obtain

$$(p-1)Q = (p-1)KT_a.$$

Since $p \geq 2$, this proves the exact identity

$$KT_a = Q = p^n. \quad (5)$$

Because T_a is an integer, (5) implies $K \mid p^n$. Every positive divisor of the prime power p^n is p^k for a unique integer k with $0 \leq k \leq n$. Finally, the definition of a bent partition includes $p \mid K$, so $k \geq 1$. Therefore $K = p^k$ with $1 \leq k \leq n$, as claimed. $\square$

Remark. The argument in fact uses only $n > 0$, not the parity of n . Thus the same conclusion holds whenever the stated Walsh-flat bent functions and the stated bent-partition property exist in odd dimension as well. Moreover, the proof gives the exact identity

$$\#\{x \in V : x \text{ and } x + a \text{ lie in the same block}\} = \frac{p^n}{K}$$

for every nonzero $a \in V$.